

Mangroves and seawalls: household choices in Semarang

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Abstract

Semarang is sinking faster than the sea is rising. This study asks how households choose between hard defences and restored mangroves, and what each choice costs them. Households near restored mangroves reported less flood damage, and their willingness to pay for protection grew once they had seen the mangroves work.

Keywords: mangroves; seawalls; land subsidence; willingness to pay; Semarang

1. Introduction

Semarang, on the north coast of Java, is sinking. Across much of the city the ground is falling faster than the sea is rising, and ordinary spring tides now reach streets that were dry a generation ago. The city has answered with a seawall and, in several neighbourhoods, with restored mangrove belts. Residents live with both, and their choices about where to live, what to repair and what to pay for tell us how adaptation works on the ground.

2. Methods

We surveyed 412 households in six coastal neighbourhoods between March and June 2021, and interviewed 24 residents, village officials and fishing groups. Flood damage was recorded for the 2020 and 2021 seasons and compared by distance to the seawall and to restored mangrove belts. We surveyed 412 households in six coastal neighbourhoods between March and June 2021.

3. Results

Households living within 500 metres of restored mangrove cover reported 38 percent less flood damage than those behind the seawall. Flood exposure was highest in the northern kampungs, where land subsidence exceeds eight centimetres a year. Land subsidence, driven by groundwater extraction, doubled the area flooded by ordinary spring tides between 2010 and 2020.

Residents valued the seawall for the certainty it gave, but most expected it to need raising again within fifteen years. Risk perception fell sharply among families who had lived behind the seawall for more than a decade.

4. Discussion

Willingness to pay for mangrove protection rose when households had seen restored belts reduce wave height. Community participation in replanting was strongest where fishing groups kept the right to harvest shellfish. These results suggest that visible protection builds the trust that paying for protection requires.

5. Conclusion

Mangroves and seawalls are not alternatives but partners: the wall buys time while the mangroves grow. Protection that households can see working earns their support, and their money.

References

- Hartmann, L. (2019). Living behind walls: coastal defence and trust. *Journal of Coastal Studies* 12, 44-61.
- Wibowo, A., Santos, M. (2020). Subsidence and flooding in northern Java. *Coastal Engineering Letters* 8, 101-118.
- Santos, M. (2021). Paying for mangroves: household valuation of green defences. *Ecological Economics Review* 33, 210-229.